



NewBear Computing Store

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Second & Third Newsletter April 1978

By including members contributions and other information the delays to the second newsletter have justified. Many apologies to all those who were beginning to believe the user group was a figment of the Bear's imagination.

Due to the spread in "Personal Computing World" 77-68's are springing up all over the place at an alarming rate. On occasions its hard to keep pace with demand particularly with edge connectors.

If any body is interested in assisting or acting as secretary to the user group, I would be very interested to hear from them. I believe that discussion is best conducted free of 'comercial' interest.

There are two other "journals" which are usually pleased to accept articles on 77-68 implementations, modifications:-

Amateur Computer Club Newsletter
7 Dordells
Basildon
Essex

and

Personal Computing World
62A Westbourne Grove
London

The second actually pays money for articles.

For those of you who have only got a photo copy of part of Design Note 22 for the soft monitor board. It now has been completed and printed. Please contact the Bear for your copy.

Introduction

The User Group has so far worked very well and has in several instances been of great help in commissioning reluctant systems. A variety of refinements to 77-68 have appeared some of which are included within these pages. Some interesting software has also arrived. Many thanks to all those who have contributed their ideas, designs and software.

The Present

Two more printed circuit boards have appeared:

1. 4K RAM board using 2102 1k static RAMS.
2. Soft monitor board with V24/RS232C or 20mA current loop.

The christmas holiday and a poisoned gold plating tank have delayed the main batch of these boards but no supply problem is expected during 1978.

The soft monitor board might need some explaining to the newcomer to computing:-

To get the best use of a general purpose Microcomputer it is necessary to attach a Visual Display Unit or Teletypewriter. Nobody can run Basic or Startreck on 7 segment display LED's. These are expensive and hard to comeby items. The interface for these items to the computer can be a serial asynchoronus port conforming to a "standard" eg V24 etc. However to organise the Computer to respond to this port a program has to be running called the "Monitor" or " Operating System". This is normally in ROM and such items as "Mikbug" can be purchased for £12 to £16. However a ROM cannot be changed without additional cost, and sooner or later the operating system will not do what you want it too. To overcome this a complete operating system can be loaded into protected RAM each time the computer is switched on via a very small program called a "bootstrap loader" through the V24 port via a Kansas City cassette interface or paper tape reader. The advantages are complete flexibility, the disadvantages that the first half minute on switch on of the computer is spent loading the operating system.

The Future

Rumour has it that several printed circuit boards are in preparation eg a Memory board using 4k dynamic RAMS which will hold either 8 or 16 kilo bytes.

An interface board that has both "Mikbug" monitoring ROM and a hardware single step feature.

A Visual Display Unit for solving the I/O problems at the lowest cost.

Unfortunately it does take several months to layout the artwork, test the prototype printed circuit boards and then take delivery of the first batch, so patience please!! The safest approach is to believe it when you see it!

"Drop offs" and "Funnies"

The hand book contains some dubious advice eg poke a piece of wire in the 6800 40 pin socket on the CPU board before inserting the 6800.

If the piece of wire is large compared to the pins of a 6800 the result is a broken socket and many hours wondering why it does not work. Many apologies to all those who have fallen into that trap.

The "78" way socket supplied has delicate pins for fitting into a "VERO" board backplane. If heavy gauge wires are soldered on without other mechanical support the pins can be snapped off. The very great advantages on an instant backplane make up for this disadvantage. The socket is in fact 80 way with the two outerpins not used. (Careful where you start counting).

Anomalous corruption of memory location 00 has been experienced by some people when using 3E WAI instruction. This is in fact not surprising if the stackpointer is not defined to somewhere out side the program area of memory. The internal registers of the 6800 are pushed into the stack during the execution of the WAI instruction and so must end up somewhere between 00 and FF in the basic machine memory.

Hardware Information

The Card frame/Racking system problem, will always be open for debate.

The cheapest will always be homemade, out of scrap using for instance double glazing plastic for card guides; oneside for the card and the other side for small screws or bolts.

The next is *Critchley Bros. Ltd., Brimscombe, Stroud, Glos. GL5 2TH. England.*
Telephone: Brimscombe 2451 (045 388 2451) Telex 43194

Order Part Nos 1406 and 5U-S.T.L.B.50
Cost £14 to £16.

The disadvantage is that there are no extras like case or hinged front panel etc.

And finally the Vero card frame which is a very elegant item in anodised Aluminium, with a range of extras if required.

(Worth the extra in my opinion as it constitutes on a very small part of the cost of the overall system)

Vero Electronics Limited
Industrial Estate
Chandlers Ford
Hampshire.

Order No 33-7111D-7266H

(ex stock from the Bear)

Hardware compatibility

In an attempt to standardise pin designation the following guide lines are offered:

1. All boards should have pin 7 cut as a slot.
2. In a card frame pin 1 should be at the bottom, 78 at the top.
3. Boards using +12v power rails should also have pin 68 as a slot.
4. The edge connector should have keyways inserted in the appropriate places.
5. Suitably isolated from the CPU board
Pin 75 to carry -12v
Pin 74 to carry +12v
Pin 73 to carry reset.

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Hardware (These come with no guarantee).

1) Interface to ETI 560 V.D.U. (not reproduced here send s.a.e. to the Bear and phone if nothing happens in 14 days).	
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STOP PRESS

- * VDU / KEYBOARD INTERFACE p.c.b available MAY / JUNE
- * Rom (2708) p.c.b on its way
- * Kansas city interface available from the bear as a stand alone unit.



AMATEUR COMPUTER CLUB

GENERAL LIBRARY

18.12.77

Frank Cato,
3 Rykneld Way,
DERBY.
DE3 7AT

Tel: Derby (0332)

513769

For starter, or sophisticate,
It's double-seven-sixty-eight.

For wage-slave, or for Potentate,
A double-seven-sixty-eight.

Computer power, served on a plate,
That's double-seven-sixty-eight.

For solid worth in solid state,
Get double-seven-sixty-eight.

f.s.k. - or Saturate,
For double-seven-sixty-eight.

Compare, branch, jump & calculate
With double-seven-sixty-eight.

Reads true, and won't prevaricate,
Trust double-seven-sixty-eight.

It's fun to build, and operate,
Is double-seven-sixty-eight.

Merry Christmas, Tim, from Cato,
And prosperous nineteen-seventy-eight-oh.



BEAR MICROCOMPUTER SYSTEMS

24 College Road, Maidenhead, Berks, SL6 6BN

HEXADECIMAL CODING FORM

PROGRAM HUNT		VERSION 1	AUTHOR KMS	DATE SENT 13.77	PAGE 1 OF 1
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
0000	4F		CLR A	EMPTY REG A	
0001	0D		SEC	SET CARRY BIT TO '1'	
0002	97FF	BACK	STA A \$FF	STORE VALUE OF A INTO LEDS	
0004	CE4000		LDR #4000	SET DELAY TIME (SMALLER = FASTER)	
0007	8D45		BSR DELAYX	GO TO DELAY SUBROUTINE	
0009	D6FF		LDA B \$FF	A SWITCH THROWN?	
000C	6031		BNE CHECK	YES GO CHECK OTHER SWITCH	
000D	49		ROL A	MOVE LIGHT LEFT ONE POSITION	
000E	20F2		BRA BACK	AND AGAIN	
0010	9D0A	CHECK	BSR DELAY50	GO DELAY 50ms FOR CONTACT BOUNCE	
0012	98FF		EOR A \$FF	CHECK WHICH SWITCH UP	
0014	D6FF	WAIT	LDA B \$FF	} WAIT UNTIL SWITCH OPENS	
0016	26FC		BNE WAIT		
0018	2D02		BSR DELAY50	CONTACT BOUNCE DELAY	
001A	20E6		BRA BACK	CONTINUE RUN	
001C	CE1000	DELAY50	LDR #1000	LOAD VALUE TO GIVE ABOUT 50ms DELAY	
001F	09	DELAYX	DEX	} COUNT DOWN UNTIL ZERO	
0020	26FD		BNE DELAYX		
0022	39		RTS	RETURN FROM SUBROUTINE	
THIS IS A "RABBIT HUNT" GAME THAT MAKES A					
LIGHT RUN AROUND. IF YOU SUCCEEDED IN THROWING					
A SWITCH UP WHEN THE "RABBIT" IS THERE THAT					
LIGHT IS EXTINGUISHED. IN CASE THE LIGHT IS NOT					
AT THE SWITCH POSITION THE GAME WILL CONTINUE					
WITH 2 LIGHTS. LIGHTS ARE ALWAYS STOPPED					
WHEN ANY OF DATA SWITCHES NOT '0', GAME CONTINUES					
WHEN SWITCHES ALL '0'. GAME ENDS WHEN ALL					
LAMPS DARK WITH SWITCHES AT '0'. TO START					
A NEW GAME THROW ONE OF THE SWITCHES					
TO '1' FOR A MOMENT.					
MEM. LOC. 05,06 DETERMINE SPEED. 0100 IS ABOUT					
AS FAST AS I CAN MANAGE BUT MAYBE YOU HAVE					
BETTER REACTIONS. GOOD HUNTING					

M.M.P. LISTING FROM 7768D

ADD	DATA	MNEMONIC
00	C6	LDAB
01	01	
02	56	RORB
03	D7	STAB
04	FF	
05	4A	DECA
06	27	BEQ
07	0B	
08	7A	DEC
09	00	
0A	20	
0B	27	BEQ
0C	03	
0D	7E	JMP
0E	00	
0F	08	
10	7E	JMP
11	00	
12	05	
13	86	LDAA
14	FF	
15	D8	EORB
16	FF	
17	7E	JMP
18	00	
19	02	

This mini-multi-purpose program can be used to show off your 7768 to the public, or play "Shoot" or even act as a flashing beacon. In the above state it will play "Shoot", i.e. when run if any of the eight data switches is high, it will exclusive or, with whatever is on the display at the time. E.G. as the display rotates if a LED is on and you make the corresponding switch (Data) high it will turn off, if the LED is low (off) it will turn on, however, if the data switch is left on then the next rotation of the display will cause error to occur. (you will soon see what I mean)

- 1) This program can also be used to rotate the display left or right at varying speeds.
E.g. to do this change the following addresses.

Add 02 becomes any of 5A/5C/56/59
14 is speed control FF and 00 are slowest.
16 becomes 30 or any unused Loc^u

Shoot can also be played at various speeds and can also be rotated in the opposite direction.

The sequencer section of the program (1) can also be produced by loading NOP's into LOC^u 15 and 16 then changing 02 to fute~~re~~ 56/59/5A/5C/

To produce the flashing beacon effect change the following:-

ADDRESS		DATA
00	-	NOP : 02
01	-	SBA : 10
19	-	: 01

To flash load FF on the data switches and run the program . See what happens if you change address 01 from SBA : 10 to NEGB : 50.

NOTES AND PLAYING INSTRUCTIONS FOR 'GUESS'

The aim of this game is to guess a two digit hex number generated randomly by the micro computer. It will help you to guess which number it is by telling you if you guess is too high, low or 'on the nose'.

To play the game, after having loaded the programme, set the data and address switches to '00', re-set and run. If the programme has been correctly loaded it will respond by displaying '88' on the L.F.D.s. Next enter a guess onto the data switches and wait. After a few seconds the micro computer should respond by displaying either '10', '10' or '00', depending upon whether you were too high, low or 'on the nose'.

If you were too high or low, after another short wait, the micro computer will ask you for another guess by displaying '88' again and so on until you get the number it was thinking of. When this has been attained, it will display '00' followed by, after a short wait, the number or tries you took to get the number. The whole process will then repeat until it is halted.

The random number generator used in this programme is the one provided by South West Technical in their software library and may be used in any programme to generate random numbers. The next number is generated from the previous one and will thus use the power up contents of memory to produce the first number, so that if you get tired of playing the same games, just change the contents of locations 'F9', FA, 'FB' and 'FC' to play a new game.

GOOD LUCK AND HAPPY GUESSING

N. K. Wright

DATAFILE HEX CODING FORM PC-1

PROGRAM	VERSION	OTHER	DATE	TIME
GUESS	#1	N.K.WRIGHT	7768	21/12/77

ADDRESS	MACHINE CODE	LABEL	OPERATOR AND OP-AND	COMMENT
10.0	4F	RERUN	CLR A	SET 'GO' COUNT TO 0
10.1	9.7.F.0		STAA F0	
10.3	2.6.0.0		LDAB #8	SET COUNTER
10.5	8.6.0.0.F.C	RPT	LDAA RNDM+3	GET M.S. BYTE OF RAND#
10.8	4.9		ASL A	SHIFT LEFT 3 TIMES TO GET
10.9	4.8		ASL A	BIT 28 (IN LINE WITH 31
10.A	4.8		ASL A	XOR WITH RND#
10.B	3.8.0.0.F.C		FOR A RNDM+3	PUT BIT 28 XOR 31 IN
10.E	4.8		ASL A	CARRY BY SHIFTING LEFT
10.F	4.8		ASL A	
11.0	7.9.0.0.F.9		ROL RNDM	ROTATE ALL FOUR BYTES
11.3	7.9.0.0.F.A		ROL RNDM+1	OF THE ENDS ROTATING THE
11.6	7.9.0.0.F.B		ROL RNDM+2	CARRY INTO THE LSB
11.9	7.9.0.0.F.C		ROL RNDM+3	THE MSB IS LOST
11.C	5.A		DEC B	DECREMENT THE COUNTER
11.D	2.6.E.0		BNE RPT	IF NOT 0 REPEAT
11.F	8.6.B.8	GUESS	LDAA #88	DISPLAY 88 THE REPRESENT
12.1	9.7.F.F		STAA FF	TOTELL YOU TO GUESS
12.3	9.6.F.F		LDAA FF	
12.5	9.1.F.F	AGAIN	CMPA FF	WAIT UNTIL THE DATA
12.7	2.7.F.C		BEG AGAIN	SWITCHES CHANGE STATE
12.9	3.6.0.5		LDAA #05	
12.B	0.8	WAIT	INX	
12.C	2.6.F.D		BNE WAIT	WAIT TO ALLOW WHOLE
12.E	4.A		DECA	GUESS TO BE POT ON
12.F	2.6.F.A		BNE WAIT	SWITCHES
13.1	4.F		CLR A	CLEAR CONDITION CODES
13.2	0.6		TAP	RES
13.3	9.6.F.9		LDAA F9	COMPARE GUESS TO RAND#
13.5	9.1.F.F		CMPA FF	
13.7	2.2.0.2		BHI TOO LOW	HIGHER. GUESS TOO LOW
13.9	2.3.1.3		BLS TOO HIGH	LOWER OR SAME. GOT IT OR TOO
13.B	8.6.0.1	TOO LOW	LDAA #01	DISPLAY 01 (TOO LOW)
13.D	9.7.F		STAA FF	
13.F	0.6.F.0	TIME	LDAB F0	
14.1	5.C		INC B	INCREMENT 'GO' COUNTER
14.2	D.7.F.0		STAB F0	



DATAFILE HEX CODING FORM

PG: 2

PROGRAM	VERSION	AUTHOR	SYSTEM	DATE
GUESS	#1	N.K. WRIGHT	7768	21/12/77

ADDRESS	MACHINE CODE	LEVEL	OPERATOR AND (P. 10)	CONTENTS
14.4	C.61051		LDA B # 05	
14.6	081	MORE	INX	
14.7	261FD1		BNE MORE	WAIT
14.9	5A1		DEC B	
14A	261FA1		BNE MORE	
14C	20D11		BRA GUESS	GO BACK TO ACCEPTING ANOTHER GUES
14E	961FF1	TOO HIGH	LDA A FF	SUBTRACT AN. 3#
15.0	061F91		LDA B F9	FROM GUESS
15.2	101		SBA	
15.3	27001		BEQ GOT IT	IF RESULT = 0 GOT IT!
15.5	8161101		LDA A #10	IF NOT DISPLAY (10) TOO HIGH
15.7	971FF1		STA A FF	
15.8	201E41		BRA TIME	GO BACK AND WAIT
15.9	861001	GOT IT	LDA A # 00	DISPLAY (00) GOT IT!
15.C	971FF1		STA A FF	
15.E	861031		LDA #03	
16.0	081	RPT #2	INX	WAIT
16.1	261FD1		BNE RPT #2	
16.3	4A1		DEC A	
16.4	261FA1		BNE RPT #2	
16.6	061F01		LDAB F0	DISPLAY NO OF TRIES
16.8	D71FF1		STA 0 FF	TO GET NO. (NOT INCLUDING WINNING TRY)
16.A	861031		LDA A #03	
16.C	081	RPT #3	INX	
16.D	261FD1		BNE RPT #3	
16.F	4A1		DEC A	WAIT
17.0	261FA1		BNE RPT #3	
17.2	2018B1		BRA RETURN	GO BACK TO GET NEW RAND #
	F01		COUNT OF MOVES TAKEN	
	F91		RANDOM #	
	FA1		" " +1	} USE START UP CONDITIONS TO PRODUCE END# OR LOAD SOMETHING INTO THEY TO CHANGE THE GAME.
	FB1		" " +2	
	FC1		" " +3	

LOONEY LANDER FOR 7768

BY J.A.DIAMOND

ADDRESS	MACHINE CODE	LABEL	MNEMONIC	COMMENTS
00	86 (00)		LDAA *00	INITIALISE
02	97 7D		STAA L.F.	FUEL(F.) 10 00
04	97 7F		STAA L.H.	HEIGHT(H.) 80 00
06	97 81		STAA L.V.	VELOCITY(V.) 80 00 (00 00)
08	86 (10)		LDAA *M.F.	
0A	97 7C		STAA M.F.	
0C	86 (80)		LDAA *M.H.	
0E	97 7E		STAA M.H.	
10	86 (80)		LDAA *M.V.	
12	97 80		STAA M.V.	
14	96 FF	NULOP	LDAA FF	IS THE ROCKET ENGINE ON OR OFF?
16	27 0A		BEQ NOTHR	
18	7D 00 7C		TST M.F.	IF THE ENGINE IS ON, IS THERE ANY FUEL
1B	26 0C		ENE THRUST	LEFT?
1D	7D 00 7D		TST L.F.	
20	26 07		BNE THRUST	
22	DE 80	NOTHR	LDX M.V.	IF THE ENGINE IS OFF
24	08		INX	INCREMENT VELOCITY
25	DF 80		STX M.V.	
27	20 0A		BRA MOD.H.	
29	DE 7C	THRUST	LDX M.F.	IF THE ENGINE IS ON
2B	09		DEX	DECREMENT FUEL
2C	DF 7C		STX M.F.	
2E	DE 80		LDX M.V.	
30	09		DEX	DECREMENT VELOCITY
31	DF 80		STX M.V.	
33	D6 80	MOD.H.	LDAB M.V.	
35	C0 80		SUBB *80	EXPOSE TRUE VELOCITY
37	24 0C		BCC PLUS	USE DIFFERENT ROUTINE FOR +VE & -VE
39	50		NEGB	VELOCITY
3A	DE 7F		ADDB L.H.	IF -VE, ADD VELOCITY TO HEIGHT AFTER
3C	D7 7F		STAB L.H.	TAKING 2'S COMPLEMENT
3E	24 03		BCC JUMP	
40	7C 00 7E		INC M.H.	
43	20 0C		BRA DISP.H	
45	96 7F	PLUS	LDAA L.H.	IF +VE SUBTRACT VELOCITY FROM HEIGHT
47	10		SBA	
48	97 7F		STAA L.H.	
4A	24 05		BCC DISP.H	
4C	7A 00 7E		DEC M.H.	
4F	27 0B		BEQ END	IF IT HAS LANDED GO TO THE END
51	96 7E	DISP.H	LDAA M.H.	IF IT HAS NOT LANDED DISPLAY MOST
53	97 FF		STAA FF	SIGNIFICANT HEIGHT
55	86 (FF)		LDAA *FF	VARIABLE TIME DELAY
57	4A	DELAY	DECA	
58	26 FD		BNE DELAY	
5A	20 B8		BRA NULOP	
5C	96 80	END	LDAA M.V.	IF YOU HAVE LANDED SLOWLY
5E	81 (83)		CMFA *83	FLASH REMAINING FUEL
60	24 12		BCC CRASH	
62	96 7C		LDAA M.F.	
64	97 FF	C.LOOP	STAA FF	DISPLAY REMAINING FUEL
66	08	XLOOP1	INX	DELAY 1 SECOND
67	26 FD		BNE XLOOP1	
69	7F FF FF		CLR FF	DISPLAY*00

ADDRESS	MACHINE CODE	LABEL	MEMONIC	COMMENTS
6C	CE (30) 00		LDA *30 00	SHORTER DELAY
6F	C9	XLOOP2	DEX	
70	26 FD		ENE XLOOP2	
72	20 F0		BRA C.LOOP	CARRY ON FLASHING FUEL
74	96 80	CRASH	LDAA M.V.	
76	80 80		SUB #80	
78	97 FF		STAA FF	IF YOU HAVE CRASHED, DISPLAY CONSTANT
7A	20 FE	INFIN	BRA INFIN	VELOCITY THEN GO INTO AN INFINITE LOOP
7C	00 00			FUEL
7E	00 00			HEIGHT
80	00 00			VELOCITY

NOTES ON THIS PROGRAMME

Bracketed figures eg.(10) are suggested values that can be changed as the player's skill develops.

*means * ie. numbers not addresses following.

HOW TO PLAY LOONEY LANDER

The object of the game is to land an imaginary space craft on the moon without crashing and using as little fuel as possible.

The space craft starts at a height set by address "0D" and at a velocity of 80 - address "11" .(+VE velocity being downwards.). The amount of fuel in the fuel tank is set by address "C9" .

All data switches are set to "0" and any one switch can be used for the accelerator. This is the only control that the astronaut (player) has. In the "1" position the engine is on and the craft is accelerating upwards burning fuel in the process. In the "0" position the engine is off and the craft is accelerating under gravity towards the moon.

The display shows the altitude of the craft and the player must land (height = 0) very slowly. This factor is set by location "5F" - 80. So the craft will have crashed if it hits the moon at 3 or more units of velocity. If the craft is crashed, a constant display of crash velocity will be seen. If the craft lands safely it will flash the remaining fuel level on the display.

If the craft runs out of fuel before landing the engine will cut out and the craft will plunge to its doom.

To replay the game, HALT then set the DATA switches to "0", RESET then RUN.

Now you can play. Good luck and happy landings.

MORE TUNE LISTINGS FOR THE "TUNER" PROGRAMME IN THE 7768 MANUAL

PAGE 1.

BY J.A. DIAMOND

GREENSLEEVES

JESU, JOY OF MAN'S DESIRING

ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE
40	74	B2	40 7B	7A	60 64	(CHANGE ADDRESS "02" TO 30)	
42	80 61	B4	CO 9C	7C	60 86	32	30 93
44	40 57	B6	BO 40	7E	90 77	34	30 83
46	60 4D	B8	10 00	80	30 70	36	30 74
48	20 49	BA	60 40	82	CO 77	38	30 61
4A	40 4D	BC	20 44	84	60 64	3A	20 6E
4C	80 57	BE	40 4D	86	60 64	3C	10 00
4E	40 67	CO	80 57	88	60 43	3E	30 6E
50	60 83	C2	40 67	8A	60 64	40	30 57
52	20 74	C4	60 83	8C	CO 59	42	20 61
54	40 67	C6	20 74	8E	BO 59	44	10 00
56	80 61	C8	40 67	90	10 00	46	30 61
58	30 74	CA	60 61	92	60 59	48	30 49
5A	10 00	CC	20 67	94	60 64	4A	30 4D
5C	60 74	CE	40 74	96	60 70	4C	30 49
5E	20 7B	DO	60 7B	98	60 64	4E	30 61
60	40 74	D2	20 8B	9A	60 59	50	30 74
62	80 67	D4	40 7B	9C	60 64	52	30 93
64	40 7B	D6	BO 74	9E	60 70	54	30 83
66	80 9C	D8	10 00	AO	60 64	56	30 74
68	40 74	DA	FF 74	A2	60 59	58	30 6E
6A	80 61	DC	FF 00	A4	60 64	5A	30 61
6C	40 57	DE	FF 00	A6	60 70	5C	30 57
6E	60 4D	EO	00 00	A8	60 77	5E	30 61
70	20 49			AA	CO 86	60	30 6E
72	40 4D	<u>ONWARD CHRISTIAN SOLDIERS</u>		AC	BO 86	62	30 74
74	80 57			AE	10 00	64	30 83
76	40 67			BO	60 97	66	30 74
78	60 83			B2	60 97	68	30 93
7A	20 74	40	60 64	B4	60 97	6A	30 9C
7C	40 67	42	60 64	B6	60 97	6C	30 93
7E	60 61	44	60 64	B8	60 97	6E	30 83
80	20 67	46	60 64	BA	30 A0	70	30 05
82	40 74	48	90 64	BC	30 B4	72	30 9C
84	60 7B	4A	30 59	BE	60 A0	74	30 83
86	20 8B	4C	CO 64	CO	60 97	76	30 6E
88	40 7B	4E	60 86	C2	60 86	78	30 74
8A	BO 74	50	60 86	C4	60 86	7A	30 83
8C	10 00	52	60 97	C6	60 86	7C	30 74
8E	FF 74	54	60 86	C8	30 97	7E	30 93
90	BO 40	56	CO 77	CA	30 86	80	30 83
92	10 00	58	30 77	CC	CO 77	82	30 74
94	60 40	5A	10 00	CE	BO 77	84	30 61
96	20 44	5C	60 97	CO	10 00	86	20 6E
98	40 4D	5E	60 77	D2	60 64	88	10 00
9A	80 57	60	60 64	D4	60 64	8A	30 6E
9C	40 67	62	60 4D	D6	60 4B	8C	30 57
9E	60 83	64	CO 4B	D8	60 50	8E	20 61
AO	20 74	66	CO 50	DA	CO 4B	90	10 00
A2	40 67	68	60 59	DC	CO 64	92	30 61
A4	80 61	6A	60 59	DE	60 70	94	30 49
A6	30 74	6C	60 77	EO	60 77	96	30 4D
A8	10 00	6E	60 64	E2	90 86	98	30 49
AA	60 74	70	CO 64	E4	30 97	9A	30 61
AC	20 7B	72	10 00	E6	CO 97	9C	30 74
AE	40 74	74	60 35	E8	CO 97	9E	30 93
BO	80 67	76	60 86	EA	FF 00		
		78	60 86	EC	00 00		

TIME LISTINGS PAGE 2.

ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE	ADDRESS	MACHINE CODE
A0	30 83	THE FOLLOWING WERE BY D.J. DIAMOND		B2	20 86	68	CO 86
A2	30 74			B4	20 8E	6A	20 77
A4	30 AF			B6	20 86	6C	20 6A
A6	30 61			B8	20 77	6E	40 64
A8	30 6E	<u>ALISHA</u>		BA	20 86	70	40 77
AA	30 74	40	80 64	BC	20 8E	72	CO 86
AC	30 83	42	20 50	BE	20 86	74	40 8E
AE	30 93	44	20 42	CO	20 6A	76	40 77
AO	30 05	46	20 59	C2	20 50	78	40 6A
B2	30 93	48	20 42	C4	20 59	7A	80 64
B4	30 9C	4A	20 64	C6	20 50	7C	40 77
B6	30 93	4 C	20 42	C8	20 64	7E	80 6A
B8	30 74	4E	20 6A	CA	20 50	80	40 64
BA	30 61	50	20 86	CC	20 6A	82	20 59
BC	30 49	52	20 8E	CE	20 50	84	20 B4
BE	30 61	54	20 86	DO	20 77	86	20 97
CO	30 74	56	20 77	D2	20 50	88	20 86
C2	90 93	58	20 86	D4	20 55	8A	20 6A
C4	FF 74	5A	20 8E	D6	20 50	8C	20 59
C6	81 6E	5C	20 86	D8	20 4B	8E	80 50
C8	FC 61	5E	20 6A	DA	20 50	90	40 55
CA	10 00	60	20 42	DC	20 59	92	80 50
CC	80 61	62	20 4B	DE	20 50	94	40 64
CE	FF 6E	64	20 42	EO	30 64	96	40 59
EO	81 74	66	20 50	E2	10 00	98	EO 6A
B2	30 83	68	20 42	E4	15 64	9A	20 A0
D4	30 05	6A	20 59	E6	15 59	9C	20 7E
D6	30 AF	6C	20 42	E8	15 64	9E	20 6A
D8	30 9C	6E	20 64	EA	40 6A	AO	80 64
DA	30 83	70	20 86	EC	40 77	A2	40 59
DC	30 93	72	20 8E	EE	80 86	A4	80 64
DE	30 83	74	20 86	FO	40 50	A6	40 77
EO	30 6E	76	20 77	F2	40 64	A8	EO 6A
E2	30 74	78	20 86	F4	80 77	AA	40 77
E4	30 6E	7A	20 8E	F6	80 59	AC	40 7E
E6	30 83	7C	20 86	F8	FF 64	AE	60 86
E8	30 9C	7E	20 64	FA	FF 00	BO	20 77
EA	30 05	80	20 77	FC	00 00	B2	20 86
EC	30 9C	82	20 86			B4	20 8E
EE	30 83	84	20 A0	I'M FOR EVER BLUWING BUBBLES		B6	80 97
FO	30 6E	86	20 CA			B8	40 77
F2	30 74	88	20 A0	40	60 86	BA	40 86
F4	30 83	8A	20 86	42	20 77	BC	CO 64
F6	30 74	8C	20 64	44	20 86	BE	40 6A
F8	60 00	8E	FF 6A	46	20 8E	CO	40 64
		90	21 6A	48	80 97	C2	40 50
		92	20 77	4A	40 77	C4	80 59
		94	20 97	4C	40 86	C6	80 77
		96	20 B4	4E	FF 64	C8	40 6A
		98	20 D6	50	41 00	CA	FF 64
		9A	20 B4	52	60 64	CC	FF 00
		9C	20 97	54	20 59	CE	FF 00
		9E	20 77	56	20 64	DO	00 00
		A0	FF 86	58	20 6A		
		A2	41 86	5A	80 64		
		A4	20 50	5C	40 77		
		A6	20 42	5E	CO 86		
		A8	20 59	60	CO 86		
		AA	20 42	62	40 77		
		AC	20 64	64	60 64		
		AE	20 42	66	20 77		
		BO	20 6A				

BCD / BINARY REACTION TIMER FOR 7768
BY J.A. DIAMOND

ADDRESS	MACHINE CODE	LABEL	MINI-MONIC	COMMENTS
00	4F		CLRA	
01	97 FF		STAA FF	CLEAR DISPLAY
03	7F 00 23		CLR 23	
06	06 05	DELAY	LDAB *05	5 SECOND DELAY
08	08	XLOOP	INX	
09	26 FD		BNE XLOOP	
0B	5A		DECB	
0C	26 FA		BNE XLOOP	
0E	C6 FA		LDAB *FA	DELAY ONE HUNDREDTH OF A SECOND
10	6D 00	LOOP	TST (7)	$250 \times \frac{25}{625000} = \frac{1}{100}$
12	B7 FF FF		STAA FF (5)	
15	B7 00 23		STAA 23 (5)	
18	02		NOF (2)	
19	5A		DECB (2)	
1A	26 F4		BNE LOOP (4)	
1C	89 01		ADCA	INCREMENT DISPLAY (BCD)
1E	19		DAA	
1F	26 ED		BNE LOOP	COUNT FOR ONE SECOND THEN STOP
21	20 FE	INFIN	BRA INFIN	END IN AN INFINITE LOOP
23	00			BYTE TO BE DISPLAYED WHEN HALTED

NOTES ON THIS PROGRAMME

The programme as it stands will count in BCD. To count in binary change address "1C" to 8B, "1E" to 02.

HOW TO USE THE REACTION TIMER

Set the address switches to 23, and the data switches to 00. RESET.
 RUN. The display will change to 00. About 5 seconds later the display will start counting. When this happens, press the HALT. The display will then show your reaction time in one hundredths of a second. To replay RESET Then RUN.



BEAR MICROCOMPUTER SYSTEMS

24 College Road, Maidenhead, Berks, SL6 6BN

HEXADECIMAL CODING FORM

PROGRAM LOADER		VERSION 2	AUTHOR CJF	DATE 15 1 78	PAGE 1 OF 1
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
0000	06FF	START	LDA B FF		
023E	00E9	RESET	LDS #RESET-1	16000E FIRST CHANGE	
0517		READY	TBA		
06CE	1700		LDA #1000	} 50 ns Delay for Switch Debounce	
0709		DELAY	DEX		
0A26	FD		BNE DELAY		
0C06	FF	READ	LDA B FF	Read Switches	
DE11			CBA	} Any change?	
DE27	FB		BEG READ		
E107	FC		SBB B TEMP	} Top bit altered?	
E398	FC		FOR A TEMP		
E52C	33		BGE NOTJET	If not, don't load yet	
E732			TSX		
E86E	00		JMP @,X	to NOTJET or LOAD depending on	
ED8E	00EE	NOTJET	LDS #LOAD-1	Have read change? Subsequent 2 loads	
ED0E	6		BRA READY		
EEDE	FA	LOAD	LOX ADDR		
F1E7	00		SBB B @,X	load switches to current ad.	
F3D7	FF		SBB B FF	and display loaded value	
F53C	00FB		INC ADDR+1	next address	
F82D	08		BRA RESET		
FC		TEMP			
FA		ADDR		Put start address in ADDR+1	
				(FB), then reset to START	

(03). Then each time the MSB of the switches is changed, program will load the value so formed into the next successive location, except that 57 may be switched first without the number being recorded, eg. to load 'F1' when the switches show '70', set them first to '71' then to 'F1'; but to load '71' from that position set 'F0', 'F1' and then '71'.

If the MFI is halted at any time, ADDR+1 will contain the next address to be loaded. If a wrong value is loaded, reset ADDR+1 to the address and reset to START.

... to the top of core so that it can

I have written this at the top of core so that it can be left there and used to load programs in earlier at will.

"QUOTE" FROM SOURCE

(ET1 - 08/77)

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(ADAPTED)

This circuit uses a 16 key calculator keyboard to generate 8-bit Binary Code, using Z Hex Digits for each 8-bit code.

If for example the code Hex 41 (ASCII 'A') is required:

'4' is pressed first. After 10 ms (to avoid switch bounce), the code for 4 is latched into the 4 M.S.B.'s of the output. '1' is then entered. After 10 ms, a 200us pulse is generated by the 2nd 74121. If ENABLE is low, a negative pulse appears at the STROBE Output, and the 8-bit ASCII 'A' appears on the other outputs. If ENABLE is high, the circuit remains in its initial state with STROBE disabled.

Sorry this is all so bitty, but it might be an idea to consult the original.

J. Messenger

131 Rushden Gardens,
Clayhall,
Ilford,
Essex, IG5 0BN
19/11/77

Dear Tim,

Sorry about the delay, but here is the details of my 77-68 VDU interface, as requested.

The VDU ENABLE line goes low when the address is between F800 and FBFF, This 1K block of memory being in the VDU circuit. This is similar to the ETI System 68 VDU. The NRDS and NWDS lines are only active when the VDU ENABLE is low.

In my system, the 77-68 memory is at FFO0-FFFF. The 256SEL line on the 77-68 is driven by the 74155 decoder. This arrangement is quite good for the following reasons:-

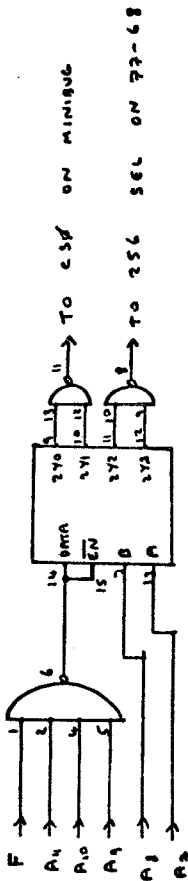
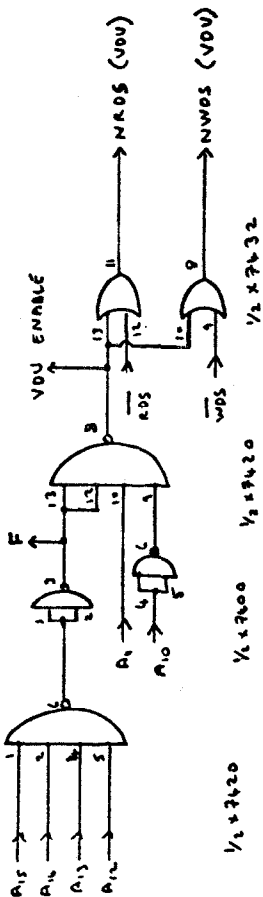
1. The output from the 74155 can also be used as the address decoder for the MINIBUG ROM, of any other memory between FEOO and FFFF in 128 byte blocks.
2. With the 77-68 memory at FFOO, the first 128 bytes can be used as the scratchpad memory for MINIBUG, without the need for a MC6810. The last few bytes can be used for the various interrupt address vectors, as well as the reset vector.

MINIBUG does have some advantages over MIKBUG. It is designed to work with a UART, which allows parallel data input at very high speed. For example, my tape reader works at 65 characters/sec which gives a fairly high baud rate. The main disadvantage, however, is the lack of a dump to tape program, which is present in MIKBUG. To over come this, I adapted the MIKBUG version to run with MINIBUG(see enclosed listing). It is resident in the undisplayed part of the VDU memory(the character generator part is like the old ETI design, but with 16 lines instead of 8, thus only using the first 512 bytes of the 1K block).

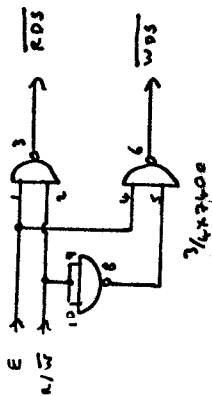
Well, that's about it. I hope it's of some use to you.

Yours sincerely,

Rick Barnett.



$1/2 \times 7420$ $1/2 \times 74155$ $1/2 \times 7400$



		NAM	PUNCH	
00001				
00002		*		
00003	FB00	ORG	EFB00	
00004		*		
00005	FB00 FF FB6D	LDX	BEGA	
00006	FB03 FF FB6F	STX	Tw	TEMP BEGINNING ADD.
00007	FB06 B6 FB72	PUN11 LDA A	ENDDA+1	
00008	FB09 B0 FB70	SUB A	Tw+1	
00009	FB0C F6 FB71	LDA B	ENDDA	
00010	FB0F F2 FB6F	SBC B	Tw	
00011	FB12 26 04	BNE	PUN22	
00012	FB14 81 10	CMP A	716	
00013	FB16 25 02	BCE	PUN23	
00014	FB18 86 0F	PUN22 LDA A	715	
00015	FB1A 8D 04	PUN23 ADD A	74	
00016	FB1C B7 FB73	STA A	MCONT	FRAME COUNT THIS RECORD
00017	FB1F 83 03	SUB A	73	
00018	FB21 B7 FB74	STA A	TEMP	BYTE COUNT THIS RECORD
00019		* PUNCH CR,LF,HULL,S,1		
00020	FB24 CF FB59	LDX	7MTAPE1	
00021	FB27 B0 FB66	JSR	PDATA1	
00022	FB2A 5F	CLR B		
00023		* PUNCH FRAME COUNT		
00024	FB2B CE FB73	LDX	7MCONT	
00025	FB2E 8D 24	ESR	PUNT2	PUNCH 2 HEX CHARACTERS
00026		* PUNCH ADDRESS		
00027	FB30 CE FB6F	LDX	7Tw	
00028	FB33 8D 1F	ESR	PUNT2	
00029	FB35 8D 10	BSR	PUNT2	
00030		* PUNCH DATA		
00031	FB37 FE FB6F	LDX	Tw	
00032	FB3A 8D 18	PUN32 BSR	PUNT2	PUNCH ONE BYTE
00033	FB3C 7A FE74	DEC	TEMP	
00034	FB3F 26 F9	BNE	PUN32	
00035	FB41 FF FB6F	STX	Tw	
00036	FB44 53	COM B		
00037	FB45 37	PSH B		
00038	FB46 30	TSX		
00039	FB47 8D 0B	BSR	PUNT2	PUNCH CHECKSUM
00040	FB49 33	PUL B		
00041	FB4A FE FB6F	LDX	Tw	
00042	FB4D 09	DEX		
00043	FB4E BC FB71	CPX	ENDDA	
00044	FB51 26 B3	BNE	PUN11	
00045	FB53 39	RTS		
00046		* PUNCH 2 HEX	UPDATE CHECKSUM	
00047	FB54 EB 00	PUNT2	ADD B X	
00048	FB56 7E FEBB	JMP	OUT2H	
00049	FB59 00	MTAPE1 FCB	FD.FA.0.0.0.0.153.131.4	
	FB5A 0A			
	FB5B 00			
	FB5C 00			
	FB5D 00			
	FB5E 00			
	FB5F 53			
	FB60 31			
	FB61 04			
00050	FB62 BD FEAE	PDATA2 JSR	OUTCH	
00051	FB63 0E	INX		
00052	FB66 A6 00	PDATA1 LDA A	X	
00053	FB68 81 04	CMP A	74	
00054	FB6A 26 F6	BNE	PDATA2	
00055	FB6C 39	RTS		
00056	FB6D 0002	BEGA	RMB	2
00057	FB6F 0002	Tw	RMB	2
00058	FB71 0002	ENDDA	RMB	2
00059	FB73 0001	MCONT	RMB	1
00060	FB74 0001	TEMP	RMB	1
00061	FB76	OUT2H	EOU	EFEBB
00062	FB78	OUTCH	EOU	EFEBB
00063		END		

Serial Interface for 77-68

The attached circuit diagram shows the breadboarded circuit I am at present using to drive a teletype for the 77-68. The system is still under development and will probably be considerably expanded before it is committed to a printed circuit board.

The circuit is based on the 1013 UART and has an RS-232 interface driving a Model 38 Teletype. Most teletypes need a 20mA current loop interface, suitable circuits are given in the ACC newsletter for April, 1977 which also describes interface standards. Full details of UART operation are given in the attached data sheets, in essence it converts characters from the parallel form of the data bus to the serial form used by the terminal, and also carries out the reverse operation. Decoding circuits are needed to make the UART look like a byte of memory to the 77-68, another byte being used for control purposes.

Address Decoding

The address decoding has been designed to allow the addition of further input/output channels with the minimum of extra circuitry.

IC3 with its associated inverters decodes a 512 byte section of memory as input/output, F600 to F7FF with the components shown. When an address in this range is detected on the address bus 256SEL is taken low to disable the bottom 256 bytes of RAM and one of the control lines (R0, W0, R1, W1) is also driven low, depending on the values of R/W and A0. The E signal is taken to IC3 to ensure this only occurs when there is a valid address on the bus. R0 and W0 allow reading or writing respectively at addresses with an A0 value of zero (F600, F602, F604..... F7FE) whilst R1 and W1 correspond to an A0 value of one (F601, F603, F605.....F7FF).

IC1 plus IC2a test bits A1 to A8 of the address bus and enable the gates of IC4 for the two addresses used for the UART, F77E for the transmission and reception of characters and F77F for control information.

UART Control

A read operation on address F77F places the following data about the UART status on the data bus:

- D0 The transmit buffer empty flag, T8MT, is set to 1 if ready to transmit another character.
- D1 1 if a parity error occurred on the last character received.
- D2 1 if a framing error occurred on the last character.
- D3 1 if an over-run error was detected on the last character.
- D4 to D6 are not used.
- D7 The data available flag, DAV, is set to 1 when a character is received.

A write operation on address F77F allows the following control information to be passed to the UART from the data bus:

- | | | |
|----|---|---|
| D0 | 1 to select even parity | |
| D1 | 0) | These control bits
are only effective
when D6 is 1. |
| D2 | 1) to select 7-bit characters | |
| D3 | 1 to select 2 stop bits | |
| D4 | 0 to select a parity bit | |
| D5 | 1 to clear DAV, allowing reception of the next character to be detected | |
| D6 | 1 to enter the control bits from D0 to D4 | |
| D7 | not used | |

Before any other operation is attempted the UART must be configured to give the required data format, this is done by setting control D6 to 1 and D0 to D4 to the appropriate values (Those given above are correct for normal teletype operation) and writing them to F77F.

IC5a, d and IC7b, c allow the control bits or DAV to be changed independantly.

Clock

IC9 is a free running oscillator at 16 times the baud rate, the components shown giving 1760Hz for 110 baud operation. The 25k pot allows adjustment of the frequency, when correct no characters will be missed or spurious ones found.

Interface Circuits

An RS-232 interface has been included, the pin numbers are given for a 25 pin "D" type socket (SKT1). To reverse the polarity of the transmitted signal it is only necessary to reverse the connections to pins 2 and 3 of IC10. If the input signal is reversed polarity, an inverter will be needed between TR1 and SI.

Adding extra Input/output channels

To derive read and write enable signals for an additional pair of channels you need add only a 74LS30 and one or two 74LS04 to decode bits A1 to A8 of the desired address, followed by a 7400 to nand the decoded address line with the four control lines R0, W0, R1 and W1.

The enable signals derived as above can then be used in conjunction with the CLK signal (if necessary) to gate data between the bus and external devices.

Software

This circuit uses polling techniques to pass data between the computer and terminals, although the UART could be used in an interrupt driven system.

Initially the control bits have to be written to F77F to generate the correct format serial data. To read data from the terminal the contents of F77F are checked until DAV (bit D7) has a value of 1, indicating that a character has been received. This character is then read from F77E, and a D5 value of 1 is written to F77F to clear DAV to zero.

To write data, F77F must be read until TBMT (bit D0) has a value of 1. The character to be transmitted can then be written to F77E.

Listings of memory load and dump programs are attached, study of these should make the technique clear. The routines in these programs correspond to the Mikbug routines of the same name.

Loader Program

This reads a program from the terminal as a string of hex characters, two per byte. Any other characters are ignored, allowing insertion of spaces and line feeds to make the print clearer. Note that all characters passed to or from the UART are ascii format and occupy bits D0 to D6 on the data bus.

The loader is put in memory via the front panel switches, starting at address 0000 and finishing at 003B. Address 0013 contains the address of first byte of the program to be loaded, 003C if it is to immediately follow the loader. Reset the computer to address 0000, release the halt switch and enter the program from the terminal. When the entire program has been loaded switch to halt, reset to the start address of the loaded program and release halt to run it.

If a further program is to be loaded address 0013 must be reset to the address at which loading is to start.

Dumper Program

This program dumps a program from memory to the terminal as a string of hex characters. Every fourth character is followed by a space, and two new line characters are given after about 64 characters.

The dumper can be loaded via the above loader, and must start at address 3C. The first address to be dumped is in 3D and last in 7 . To run the dumper reset to 5D and release the halt switch. If another run is required 3D must be set to the first address to be dumped again.

If you would like to discuss any of the above give me a ring.

C.E.BRICKELL
01-937-2115

0	0000	0	S	1000	128	5	D destination addressing	H immediate address
1	0001	16	Y	1001	144	7	PC branch instruction	X indirect address
2	0010	32	A	1010	160	10	R register address	C "C" as in ASCII
3	0011	48	B	1011	176	11	IF PC < D	literal address
4	0100	64	C	1100	192	12	R = D - (Z + PC)	"N" is a hex number
5	0101	80	D	1101	208	13	IF PC > D	"N" is an octal number
6	0110	96	E	1110	224	14	R = D + 254 - PC	"N" is a binary number
7	0111	112	F	1111	240	15		"N" is a decimal number

PROGRAM		DUMPER		PAGE 1 of 1	
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
3C	9600	PUT2H	LDA A \$00	} Load acc. A with next byte and put on stack	
3E	7C003D		INC \$003D		
41	36		PSH A	} Right justify & output 1st nybble	
42	44		LSR A		
43	44		LSR A		
44	44		LSR A		
45	44		LSR A		
46	8D01		BSR PUTHR	} Return full byte to acc. A	
48	32		PUL A		
49	840F	PUTHR	AND A # \$0F	} Convert right nybble to hex.	
4B	8B30		ADD A # \$30		
4D	8139		CMP A # \$39		
4F	2302		BLS PUTCH		
51	8B07		ADD A # \$07	} Read UART status until TBMT=1	
53	F6F77F	PUTCH	LDA B \$F77F		
56	56		RFR B	} Write character from acc. A	
57	24FA		BCC PUTCH		
59	B7F77E		STA A \$F77E	} Output 4 hex characters + space	
5C	39		RTS		
5D	3DD	PUT4HS	BSR PUT2H	} Output 4 hex characters + space	
5F	8DD	PUT2HS	BSR PUT2H		
61	8620		LDA A # \$20	} Has the 16th byte been output?	
63	8DEE		BSR PUTCH		
65	963D		LDA A \$3D	} Has the 16th byte been output?	
67	8AE1		ORA A # \$E1		
69	81FF		CMP A # \$FF	} New lines	
6B	2606		BNE TESTEND		
6D	860A		LDA A # \$0A	} New lines	
6F	8DE2		BSR PUTCH		
71	8DE0		BSR PUTCH	} Has end address been reached?	
73	963D	TESTEND	LDA A \$3D		
75	81FE		CMP A \$FE	} Has end address been reached?	
77	23E4		BLS PUT4HS		
79	3E		WAI		

DL 77-68 De bugging Information

If your P.C.B. fails to respond to the tests with 6800 and memory missing as described at the bottom of p.11 and top of p.12; The following check out procedure may help in locating the fault.

Use a df. voltmeter for these tests.

1) To check all address switches are working.

e.g.	A0	X9, 3
	1	6
	2	13
	3	10
	4	X8 3
	5	6
	6	13
	7	10

When switch is down 0 volts. = '0'

When switch is up +5 volts. = '1'

(Halt switch is set e.g. X 13 pin 2 = 0 volts. for all those tests)

If all is well procede to (2) otherwise look for short or open circuit in switch wiring or P.C.B.

2) To Test X8, X9

Halt switch set

A0	X9, 4	A4	X8, 4	Test switches for 0, 2, and 5V as in (1).
1	7	A5	7	
2	12	A6	12	
3	9	A7	9	

Set all address sw. to '0'

Test	A0	X9, 4	A4	X8, 4
	A1	7	A5	7
	A2	12	A6	12
	A3	9	A7	9

for moving halt switch from set to RUN.

This should result in 0 Volts Halt set
3.8 Volts Halts Switch to RUN.

If this is not the case suspect

a) ADSEL

b) Short circuit in X13 address wires.

If O.K. procede to(3)

3) Input of X7

Halt sw set

A0	X7, 6	A4	X7, 2
1	5	5	1
2	4	6	11
3	3	7	12

Move address switches and check signal reaches the pins X7, 8. When all switches up this should go to 0 volts. Otherwise 3.8 volts.

4) To Test SWSEL

X19, 13 should go from 3.8v to 0v when the last of the address switches is set high. If this is the case procede to 5) otherwise;

X3, d, 4, 5, should go to 0v from 3.8v when last address switch is set high.
X3, 10, 11 should be 3.8v

X19, 2 should go to 0v from 3.6v when last address switch is set high.
X19, 3 should be 0v

X19, 1 should go from 0v to +3.6volts when last address switch is set high.

X19, 12 should be 0v going to +3.6volts when 'Load' is pressed.

X19, 11 should be 0v going to +3.6v when the last address switch is set high.

X19, 13 should be 3.8v going to 0v when last address is set high.

SWSEL Works. Procede to(5)

5) To Test Data Switches and Buffers

D0	X23, 2	D4	X22, 2
1	12	5	12
2	5	6	5
3	9	7	9

0 volts for switch down = '0'
+5 volts for switch up = '1'

SWSEL	X22, 10,	X23, 10,
	4	4
	13	13
	1	1

Set all but one address switch high.

Setting the last one should cause these pins to go from 3,7 to 0

D0	X23, 3	D4	X22, 3
D1	11	D5	11
D2	6	D6	6
D3	8	D7	8

Should all be 1.7 volts indicating a floating input to the optinal buffers and high impedance o/p state of X22, X23, when the address switches are not in the all high position.

With the address switch all high check these pins following the data switches.

6) To Test Data LED's and X20, X21

With Halt switch set and all address switch high.

D0	X20, 7	D4	X21, 7
1	6	5	6
2	3	6	3
3	2	7	2

and pins X20, 4, 13 (RSEL)
X21, 4, 13 are + 0.5v dropping to zero when Halt switch is set to run.

Test each data switch and appropriate X20, X21 i/p pins as above.

o/p of X20, X21.

D0	X20, 8	D4	X21, 8
1	11	5	11
2	14	6	14
3	1	7	1

Should be 0 volts when Data is High (up)
4 volts when Data is Lo.

Voltage on LED's should be 4.1 off
3.3 on approx.

It all works!

8) Quick LED Check

If address switch are low the inputs to X20, 21 float high and LED'S should all light up.